

MERIDIAN: A Synthetic-Society Crisis Simulation Engine — Execution-Ready Plan

Working codename: MERIDIAN (project-level; distinct from any in-scenario names). **Status:** Independent R&D concept, unrelated to any existing commercial platform. **Nature of the deliverable:** fictional simulation and training software — not a predictive intelligence tool, not a real-world forecasting system, and not built on or targeting real operational vulnerabilities.

Executive Summary

MERIDIAN is a synthetic-society simulation engine with a multiplayer government-strategy game layered on top of it. The engine models a national population across three tiers — macro (national indicators), meso (weighted cohorts and institutions), and micro (individually simulated key actors) — using deterministic rules, seeded stochastic draws, agent-based modelling, and network-diffusion models as the primary computational substrate. A large language model is used selectively: to interpret player decisions, propose actions for micro-tier agents, generate advisor dialogue and briefings, and compose adversarial-campaign content — but it never directly mutates macro or meso numeric state. That boundary, enforced in code rather than merely in documentation, is the single most important architectural commitment in this plan.

The initial playable demo is a fictional small island/strait state ("Republic of Vantara") facing a grey-zone maritime crisis, but the engine, schemas, and UI are archetype-agnostic from day one: additional nation types (oil-rich state, landlocked resource-poor state, regional mid-power, global superpower, failed state, federated state) are added as scenario-configuration data, not new code.

This document sets out, in order: the competitive/prior-art landscape and what is directly reusable; a full open-source licensing audit with an approved/flagged/reject classification; the three-tier simulation design and determinism boundary; the demo scenario specification; the UX and visual system; the recommended technical architecture and stack; the AI-assisted development workflow and governance model (including what "UltraCode" actually is); the security and threat model; the nation-type expansion framework; and a phased build roadmap. A runnable starter repository scaffold accompanies this document, implementing the determinism and LLM-isolation boundary in code, with a passing test suite.

Throughout, claims are labelled: **Verified fact** (cited to a primary/authoritative source), **Reasonable inference** (synthesis grounded in cited facts), **Assumption** (a starting premise not independently verified), **Unknown** (a gap the research could not close), **Requires legal advice**, or **Requires technical experiment**.

1. Similar-System Landscape: What Already Exists and What Is Reusable

Fifteen categories of prior art were surveyed — academic generative-agent research, commercial and open-source agent-based modelling (ABM) frameworks, professional wargaming platforms, digital twins, central-bank agent-based economic models, and commercial crisis-simulation products. The full findings, with per-system architecture, licensing, reproducibility, and LLM-usage assessments, are preserved in the research appendix; the operative conclusions are below.

1.1 Nothing existing does the whole job — but five categories are directly reusable

No single system combines a census-grounded synthetic population, a deterministic rules/diffusion core, and a selectively-invoked LLM agentic layer, wrapped in a multiplayer government-decision-maker game. **Reasonable inference**, grounded in the verified facts about each individual system below. The closest single-system analogs are narrow: DeepMind's **Concordia** provides a production-grade "Game Master" pattern for grounding LLM agents ([Concordia GitHub](#)); CAMEL-AI's OASIS scales LLM-agent populations to roughly one million users on a simulated social-media substrate ([OASIS GitHub](#)); and Conducttr is the closest commercial analog for the human-decision-maker-plus-simulated-information-environment experience, used by NATO's Allied Rapid Reaction Corps ([Conducttr DSEI page](#)). None of these on its own satisfies the brief; the correct approach is to compose reusable layers.

Natural build-stack mapping, synthesizing the landscape research:

1. **Synthetic population layer (the "who")** — PopGen/SynthPop-style census-grounded generation via Iterative Proportional Updating (IPF/IPU), which is deterministic, open-source, and demographically defensible ([PopGen3 GitHub](#); [SynthPop GitHub](#)).
2. **Mechanistic reaction layer (fast, deterministic-seedable)** — NDlib or the faster CyNetDiff for information diffusion (Independent Cascade / Linear Threshold / SIR) over a social graph; Mesa for spatial and institutional agent-based modelling ([NDlib GitHub](#); [Mesa docs](#)).
3. **Generative/agentic layer (believable individual reactions)** — Concordia's Game-Master pattern for the government-decision-maker adjudication loop, and CAMEL/OASIS's societies-and-recommendation-system pattern for population-scale information-warfare dynamics ([Concordia](#); [OASIS](#)).
4. **Orchestration and human-in-the-loop** — Microsoft's Agent Framework (the April 2026 production unification of AutoGen and Semantic Kernel) for role orchestration, with the human government decision-maker as a human-in-the-loop participant ([Microsoft Agent Framework guide](#)).
5. **Domain and doctrine references** — the NATO Wargaming Handbook's design→develop→execute→analyse process, NATO's SIR-based disinformation "Synthetic Environment" model, and RAND's experimental-wargaming methodology, none of which are shippable software but all of which inform the design ([NATO Wargaming Handbook](#); [NATO SEN counter-disinformation study](#)).

1.2 Systems assessed as low relevance despite surface similarity

DeepMind's Genie/SIMA world-model family generates photorealistic, navigable 3D environments, but DeepMind itself lists "difficulty modeling interactions among multiple independent agents" as an explicit Genie 3 limitation, and the interaction window is only "a few minutes" — the opposite of what a multi-week social-institutional simulation needs ([Tech Times on Genie 3 limitations](#)). It is noted only as a possible future visualization layer, not a core system.

Government/military constructive-simulation platforms (MASA SWORD, LLNL's JCATS) are mature, fielded, AI-driven, and government-controlled, but are proprietary, combat-force-focused, and carry no civilian synthetic-population or social/institutional-graph layer ([MASA SWORD v6.0](#); [LLNL JCATS](#)). They are useful only as a fidelity/credibility benchmark, not as reusable code.

Virtual Singapore / Dassault 3DEXPERIENCity and Improbable Defence's Skyrall are relevant as references for large-scale, real-time synthetic-environment architecture, but both are proprietary and Improbable's defence unit was reported in October 2025 to be in talks to divest following a US office closure — a live sustainability risk for anyone considering it as a dependency ([Shephard Media on Improbable divestiture](#)).

1.3 Central-bank and epidemiological ABM precedent for the "engine, not vibes" argument

Central banks including the Bank of England have used heterogeneous-agent ABMs "to understand the economy from the bottom up," where macro outcomes emerge from micro interactions rather than being authored top-down — directly analogous to MERIDIAN's macro state emerging from meso/micro dynamics rather than from LLM narration ([Bank of England 2016 Quarterly Bulletin](#); [BoE 2025 ABM review](#)). Nation-scale epidemiological ABMs such as Epicast 2.0 (roughly 324 million agents for the entire US population) prove that the synthetic-population-plus-contact-network-plus-contagion-over-network pattern is HPC-proven and directly transferable to modelling crisis diffusion (panic, rumor, disruption cascades) at population scale ([Epicast 2.0](#), [arXiv](#)).

1.4 Licence-friendliness of the landscape, at a glance

Permissive and directly reusable: Mesa (Apache-2.0), Concordia (Apache-2.0), CAMEL/OASIS (Apache-2.0), Microsoft AutoGen/Agent Framework (MIT), SynthPop (BSD-3-Clause), NDlib (BSD-2-Clause). Copyleft or otherwise constrained: NetLogo (GPL-2.0-or-later), GAMA (GPL-3.0). Proprietary/closed and not reusable as code: MASA SWORD, JCATS, Virtual Singapore/3DEXPERIENCity, Improbable Skyrall, Conducttr, Genie. Full per-system detail (intended users, architecture, reproducibility, maintenance status, and sourcing) is preserved in the underlying research file for reference during implementation.

2. Open-Source Licensing Audit

Scope and method note (verified fact about this audit's own methodology): every licence claim below was checked against the actual licence file in the repository, not the README, consistent with the brief's explicit instruction that "open source" alone is not sufficient. This is a technical risk assessment, not legal advice — items marked "Legal review required" must go to qualified counsel before production use. The audit assumes the reference product model is a proprietary, closed-source, multi-tenant SaaS sold to enterprise and government customers, which is the condition that makes AGPL, SSPL, and Business Source Licence (BSL/BUSL) "as-a-service" clauses material.

2.1 Summary table — approved components

Component	Licence (SPDX)	Commercial use	Copyleft/network trigger	Recommendation
Mesa (Python ABM)	Apache-2.0	Yes, unrestricted	None	Approved
NetworkX	BSD-3-Clause	Yes, unrestricted	None	Approved
Concordia	Apache-2.0	Yes, unrestricted	None	Approved
AutoGen / Microsoft Agent Framework	MIT (code) + CC-BY-4.0 (docs)	Yes, unrestricted	None	Approved
LangGraph, LangChain, CrewAI	MIT	Yes, unrestricted	None	Approved
Deck.gl , Sigma.js , Cytoscape.js , D3.js	MIT / ISC	Yes, unrestricted	None	Approved
MapLibre GL JS	BSD-3-Clause	Yes, unrestricted	None	Approved (MapLibre is the Mapbox v1.x community fork prior to Mapbox's proprietary relicensing)
CesiumJS (core)	Apache-2.0	Yes, unrestricted	None	Approved — Cesium ion (hosting) is a separate paid SaaS
NDlib	BSD-2-Clause	Yes, unrestricted	None	Approved
Apache Kafka	Apache-2.0	Yes, unrestricted	None	Approved (permissive alternative to Redpanda)
Temporal (self-hosted server)	MIT	Yes, unrestricted	None	Approved — Temporal Cloud is separate paid SaaS
Ollama, llama.cpp	MIT	Yes, unrestricted	None	Approved — runtime only; model weights carry separate licences
PostgreSQL	PostgreSQL Licence (BSD/MIT-like)	Yes, unrestricted	None	Approved — "gold-standard permissive DB licence"

Component	Licence (SPDX)	Commercial use	Copyleft/network trigger	Recommendation
pgvector	PostgreSQL Licence	Yes, unrestricted	None	Approved

(Full audit source: [research_licensing.md](#) summary table, verified against each project's raw LICENSE file on 18 July 2026.)

2.2 Flagged non-permissive or restrictive licences

Licence class	Components affected	Core commercial risk
GPL-3.0 (strong copyleft)	Neo4j Community Edition	Linking/embedding forces the combined work under GPL. Safe only run as a separate process over the Bolt network protocol — never link/embed (Neo4j licensing).
AGPL-3.0 (network copyleft, historical)	Neo4j's earlier AGPL lineage	If an AGPL Neo4j code path is present, network users must be offered corresponding source — fatal for closed SaaS. Confirm the deployed edition is GPLv3 Community Edition, not AGPL.
Business Source Licence (BUSL/BSL 1.1) — not OSI open source	Memgraph, ArangoDB, Redpanda	Additional Use Grants specifically bar database-as-a-service/streaming-as-a-service and building a competing offering. High risk if exposed to customers as part of the service. Converts to a permissive licence 4 years after each release, but that is not "now." Legal review required.
Source-available vendor licence (Timescale Licence)	TimescaleDB <code>ts1/</code> features only	TSL features (continuous aggregates, compression) cannot be offered as DBaaS/SaaS. Apache-licensed core features are fine if <code>ts1</code> binaries are disabled.
Proprietary Terms of Service	Mapbox GL JS v2.0+	Requires a paid Mapbox account; not redistributable as open source. v1.x was BSD-3 — this is why MapLibre (the v1.x fork) is recommended instead.
Custom "community"/open-weight model licence	Llama family model weights	Commercial-friendly but with an Acceptable Use Policy, mandatory "Built with Llama" attribution, a "Llama" naming prefix requirement on derivatives, a 700-million-monthly-active-user commercial trigger, and an IP-litigation termination clause. Approved with conditions.
Research/non-commercial model licences (MRL, MNPL)	Certain Mistral models (e.g. Codestral, some Ministral variants)	Non-commercial only — prohibited in a revenue-generating SaaS unless a paid Mistral commercial licence is separately obtained. Must be checked per model , not per vendor. Legal review required per model.

No SSPL, Elastic License 2.0, Commons Clause, or PolyForm licences were found among the audited components as of the audit date ([research_licensing.md](#)).

2.3 Recommended low-risk default stack

Standing on Postgres as a single spine — relational storage, PostGIS for geospatial, TimescaleDB (Apache core only) for time series, pgvector for retrieval, and Apache AGE for graph queries — keeps the number of licence regimes to audit and the number of systems to secure/back up/deploy into a future air-gapped environment to a minimum. Escalate to Neo4j Community Edition (run only as an isolated separate process) only if graph-query needs genuinely outgrow

Apache AGE, and treat that escalation as requiring a documented legal sign-off given the GPL-3.0 trigger. Kùzu is explicitly **not recommended** despite a clean MIT licence: its creator abandoned the project in October 2025 after Apple's acquisition, which is a "do not adopt for a multi-year product" signal regardless of licence terms ([The Register on Kùzu abandonment](#)).

3. Simulation Design: Three-Tier Architecture and the Determinism Boundary

This is the load-bearing design decision of the entire plan, directly answering the brief's core requirement that the simulation "must not merely be an LLM improvising a story."

3.1 The three tiers

MACRO tier (national/system state) is fully deterministic-plus-seeded-stochastic, computed once per simulation tick by rules and Monte Carlo draws, with zero LLM involvement in the numbers. It tracks indicators such as inflation, unemployment, GDP growth, shipping throughput, government approval, per-institution trust, military readiness, alliance confidence with each bloc, a social-stability index, foreign direct investment flow, fuel reserve days, and public-finance figures (treasury reserve, deficit, debt-service ratio).

MESO tier (cohorts and institutions) uses weighted representative agents, where one agent record represents many thousands of real citizens sharing a demographic, belief, and media-exposure profile — explicitly avoiding "every citizen a continuously running LLM," per the brief's mandate. Each cohort record carries demographics, an economic profile with a quantified sensitivity to specific shocks (e.g. income sensitivity to shipping disruption), media-exposure fractions across channel types, belief scores, an explicit grievances list, influence-susceptibility scores broken out by appeal type (authority, identity, economic), and network-position data (internal cohesion, bridging ties to other cohorts). Cohort belief updates run through a probabilistic diffusion model over the social/institutional graph; the LLM is used only to generate a natural-language label for a shift the diffusion model already computed — never to compute the shift itself.

MICRO tier (individually simulated agents — ministers, commanders, key adversary figures) carries the fullest structured state: prioritized objectives, beliefs, personality/behavioural traits (risk tolerance, status-seeking, institutional trust, corruption susceptibility), controlled resources, signed relationship scores to other named agents, an information-access list defining what that agent can legitimately know, hard legal/procedural constraints, and a decaying event memory. The LLM's role here is precisely scoped: reasoning about which action a given agent would plausibly choose given its state, generating that agent's dialogue and narrative text, and interpreting ambiguous player input into structured form. The simulation engine — not the LLM — independently validates whether a proposed action is legal and feasible, and it is the engine's deterministic rules that compute the numeric consequences.

3.2 The determinism boundary table

Layer	Computation mechanism	LLM involvement
Macro indicators	Deterministic rules + seeded Monte Carlo draws	None
Cohort belief updates	Probabilistic diffusion model over the social graph (threshold/SIR-style)	None for the numbers; may label the resulting shift in a briefing sentence
Institutional agent decisions	LLM proposes an action from structured state; the engine independently validates legality/feasibility and computes cost/effect	Proposes only — engine decides validity and magnitude
Adversary campaign design	LLM composes campaign content from a fixed schema (channel choice, messenger framing); the diffusion model computes whether and how far it actually spreads	Creative content only; engine computes numeric effect
Advisor dialogue/briefings	Fully LLM-generated, but grounded by retrieval over the current true state — not free hallucination	Primary generator, constrained to cite real state fields
Player decision interpretation	LLM parses free text or a composed decision into a structured, schema-validated intervention object	Primary interpreter; the engine rejects invalid or out-of-scope actions

This boundary is enforced in the accompanying scaffold's code, not only in this document. The LLM gateway module's function signatures return proposal objects with no access to the macro-state object at all; only the engine's tick-processing function can write to macro state, and it deliberately treats the proposal's own numeric fields as advisory rather than authoritative. A dedicated test in the scaffold (`test_llm_gateway_cannot_write_state`) exists specifically to guard this property against regression.

3.3 Reproducibility mechanics

Every simulation run has a seed; given the same seed, the same scenario version, and the same sequence of player decisions, macro and meso state is exactly reproducible — verified in the scaffold by a test that runs two identically-seeded 20-tick simulations and asserts byte-identical resulting state, alongside a companion test proving that different seeds produce divergent outcomes (**verified fact about the scaffold's own test suite**, confirmed by independently re-running `pytest` — 5 of 5 tests pass). LLM outputs — dialogue, flavour text — are logged verbatim and versioned by model ID, prompt version, and temperature, but are explicitly not part of the reproducible numeric state; they sit as an interpretive layer on top of it. Every tick produces an immutable state snapshot plus an event-log delta, which enables scenario branching (forking a new run from any historical snapshot) and Monte Carlo batch analysis (many runs from the same seed family to build an outcome-distribution baseline for after-action comparison).

The explanation system answers "why did X change" by walking the causal chain recorded in each event's `causal_parents` field back through the diffusion model and rules engine — this trace is data produced by the deterministic layers, not an LLM-generated justification, though an LLM may be used afterward purely to phrase the trace in natural language for a human reader.

3.4 Supporting schemas

Beyond the three tier-defining schemas above, five supporting object types complete the data model — full JSON Schema definitions ship in the scaffold's /schemas directory:

- **Relationship:** signed edges in the social/institutional graph — valence, trust, dependency, and last-interaction tick between any two agents.
- **Event:** the causal-chain unit — tick, type, actors involved, visibility level (public/classified/leaked), causal parents, and effects — this is what the explanation system walks.
- **Intervention:** a validated player-action object — actor role, action type, target, resource cost, a legality check result, timeline, and any second-order hooks it triggers.
- **Narrative:** a claim's lifecycle — originating campaign (if any), ground-truth status, and per-cohort adoption and resistance levels.
- **Outcome:** the result of a decision — macro deltas, meso deltas, individual micro reactions, an explanation trace, and a confidence score.

3.5 Information-environment and adversarial-campaign model

The campaign schema captures a sponsor, a strategic objective, targeted cohorts, the grievance being exploited, the narrative claim and its actual truth status, the messenger persona and its perceived independence, distribution channels, an amplification-network description (coordinated accounts, sympathetic commentators, estimated reach), a trigger event, a desired behavioural outcome, and live metrics including exposure, belief adoption, resistance, detection probability, and attribution probability. Critically, campaign *effectiveness* is computed by the diffusion model against each targeted cohort's influence-susceptibility and network-position fields — the LLM's job is limited to composing plausible campaign content within this fixed schema; it never decides how far the campaign actually spreads.

This schema was independently found to map almost one-to-one onto the **DISARM Red Framework**, the leading open, MITRE ATT&CK-style taxonomy for describing real influence operations, maintained by the DISARM Foundation and built with MITRE, FIU, and CogSec Collab ([DISARM Foundation framework repository](#)). DISARM structures a campaign as a four-phase kill chain — **Plan** → **Prepare** → **Execute** → **Assess** — and supplies named tactics for each schema element: Target Audience Analysis and Microtargeting for audience segmentation, Develop Narratives (including "leverage existing narratives" and "seed kernel of truth") for the grievance/narrative link, Establish Assets/Legitimacy for the messenger, Select Channels and Affordances for distribution, and Maximise Exposure/Conduct Pump Priming for amplification ([DISARM Red framework tactics list](#)). Adopting DISARM's vocabulary, rather than an ad hoc schema, gives the campaign model a credible, externally-recognized foundation and a path to STIX2-based threat-intelligence interoperability if the product later needs it ([DISINFOX interoperable disinformation representation, arXiv](#)).

The diffusion mechanics underneath the campaign model rest on well-established computational social science: the **Independent Cascade** and **Linear Threshold** models of Kempe, Kleinberg and Tardos (2003) are the two workhorse influence-maximisation models,

and their monotone-submodular influence function is what makes principled seed-selection tractable ([summary citing Kempe, Kleinberg & Tardos 2003](#)). Centola and Macy's distinction between simple and complex contagion matters directly for campaign design: simple contagions (like a single viral post) spread via long weak ties, but complex contagions — which require reinforcement from multiple sources before a person adopts a belief — depend on "wide bridges" between overlapping clusters, and excessive network homophily can fragment a population into echo chambers that block cross-cluster spread entirely ([Centola & Macy 2007](#); [Centola 2015, American Journal of Sociology](#)). Coordinated-inauthentic-behaviour detection research — the same literature analysts use to detect real bot networks — supplies a natural signal set (synchronized posting windows, near-duplicate content, anomalously high eigenvector centrality) that can be simulated on the adversary side and tested against on the detection side ([arXiv survey on coordinated online behaviour](#)).

On counter-messaging: the classic "backfire effect" (that correcting misinformation increases belief in it) has been substantially undermined by later, better-powered replication work, and a PNAS synthesis concludes backfire effects do not explain the durability of political misperceptions, recommending that corrections target the identity-narrative linkage and elite cue-givers rather than assuming corrections are counterproductive ([PNAS on the durability of misperceptions](#)). **Reasonable inference:** the simulation should model counter-messaging as usually net-corrective by default, with a parameterized, context-dependent backfire term reserved for edge cases such as strong prior identity attachment — rather than defaulting to the folk-psychology assumption that corrections backfire.

4. Demo Scenario: "Kestral Strait" (Republic of Vantara)

Entirely fictional; no real operational vulnerabilities, real organisations, or real named individuals are used, per the brief's explicit constraint.

4.1 Country profile

The Republic of Vantara is a 4.1-million-person archipelago of 340 islands controlling the Kestral Strait, a fictional chokepoint carrying roughly 9% of a fictional region's containerised trade. Population composition: 61% Vantaran-majority (coastal/urban), 22% Northern Creole minority (coastal fishing communities, historically economically marginalised), 11% inland highland minority, 6% foreign-resident/diaspora-linked. The economy (~\$38B GDP) is trade-dependent on shipping transit fees, fisheries, light manufacturing, and tourism, with a currency loosely pegged to a basket including a Western reserve currency. Politically it is a unitary parliamentary republic with a fragile 54%-of-seats coalition government, a free press with some state-linked outlets, and a moderate-capacity independent judiciary. Legally, defence procurement above \$10M requires competitive tender with a mandatory 18-month minimum cycle, and emergency powers require parliamentary renewal every 90 days.

Militarily, Vantara fields six ageing offshore patrol vessels (average age 27 years, two currently non-operational), no fast-attack craft, 14 mixed-maintenance-status aircraft, a small 1,800-person marine corps, and coastal radar with known gaps in the southern strait — sustaining roughly 40% continuous patrol coverage. Its small domestic intelligence service has limited signals capability, relies on a Western ally for satellite/SIGINT sharing, and has no independent

cyber-attribution capability. Foreign relationships split between a conditional, non-treaty Western Bloc security relationship (aid tied to human-rights and procurement-transparency benchmarks) and a growing, faster, fewer-conditions but longer-term-dependency-risk Eastern Bloc equipment relationship. The neighbouring failed state, "Ostrana," collapsed six years ago into three competing armed factions, one of which — the "Free Coast Movement" — has documented links to piracy and is suspected but unconfirmed of covert support from a third-party regional spoiler power.

4.2 Initial crisis state

Three commercial vessels have been attacked or boarded in the strait over the preceding six weeks, with one crew fatality and roughly \$40M in cargo losses. Shipping insurers have raised strait transit premiums 340%, and two major carriers have publicly threatened rerouting, which would add roughly nine days and substantial cost to regional trade. Domestic media narrative is fracturing: state-linked outlets blame Ostranan factions, two independent outlets allege government negligence, and a fast-growing short-video narrative of unclear origin — later revealed as a regional-spoiler-power-run operation (camp-eastern-veterans-01) — blames Western naval presence for "provoking" the attacks. Government approval sits at 44% and falling, with the opposition calling for an emergency debate. The coastal Creole fishing cohort (representing 14,200 people) is losing income to informal patrol-driven exclusion zones, and a pre-existing capital-elite-neglect grievance in that cohort is being actively targeted by the adversarial messaging campaign.

4.3 Hidden adversarial activity (not initially visible to the player)

The regional spoiler power is running campaign camp-eastern-veterans-01 — a fronted "veterans' association" account cluster amplifying a "Western provocation" narrative, targeted at the nationalist-youth and coastal-Creole cohorts, with a scheduled trigger event (arrival of a Western naval liaison vessel) planned for tick+3. Attribution probability starts very low (0.05) and rises only as the player's intelligence lead invests resources in investigation — deliberately modelling the real-world asymmetry between suspecting and proving foreign interference.

4.4 Player roles and asymmetric private information

Six roles — Head of Government, Defence Chief, Finance/Central Bank, Foreign Affairs, Intelligence Lead, and Strategic Communications — each start with distinct private information and a distinct core conflicting objective (for example, the Defence Chief privately knows vessel readiness is worse than public statements suggest and wants immediate procurement acceleration, while Finance privately knows the true treasury-reserve runway and will block unfunded defence asks). This asymmetric-information structure is the mechanical core of the multiplayer game, not a cosmetic detail.

4.5 Strategic paths, adversarial strategies, and second-order consequences

Three illustrative (non-exhaustive) strategic paths are defined — Western alignment, Eastern alignment, and a non-aligned/multilateral path — each with explicit trade-offs rather than a "correct" answer. Three adversarial strategies are defined for the regional spoiler power: narrative escalation timed to Western naval visits, attack-tempo modulation to manage attribution risk, and a slow-burn elite co-optation subplot. Twelve worked second-order consequences are specified as concrete engine outputs the simulation should be able to produce — for example: fishing-exclusion zones reduce Creole cohort income, which raises grievance, which raises susceptibility to the adversarial narrative, which raises protest probability; or accepting Eastern Bloc equipment creates spare-parts dependency that becomes a future leverage point in an unrelated negotiation. Win, loss, and — most pedagogically valuable — ambiguous outcome conditions are defined explicitly, with the explanation system required to surface trade-offs (for example, a contained crisis at the cost of deep bloc dependency or eroded civil liberties) rather than collapsing everything into a simple win/loss score.

5. UX and Visual System

5.1 Original visual language

The brief explicitly prohibits copying the trade dress of Palantir, Anduril, Call of Duty, Battlefield, Halo, Rainbow Six, or Ghost Recon — no borrowed logos, fictional-organisation branding, icon packs with restrictive terms, typefaces, sounds, or distinctive layouts. The recommended direction draws only on the general, non-proprietary territory of modern operations-centre and intelligence-fusion software conventions (dark command-console aesthetics, monospaced telemetry, status-colour discipline) without referencing any named product.

Design tokens: a graphite/near-black six-step surface scale (#0B0E11 to #151A1F), muted military-green for confirmed/nominal states (#4C6B4F), slate for neutral UI (#5C6B78), and sparing status accents — amber for caution/pending, red for critical/hostile, cyan for friendly/verified-intelligence — with never more than one accent hue dominant per screen. Typography pairs an open-licence condensed grotesk (e.g. Archivo Condensed or Barlow Condensed) for headers with an open-licence monospace (e.g. JetBrains Mono or IBM Plex Mono) for telemetry, coordinates, and numeric readouts. An 8px spacing unit and 12-column grid underlie a fixed sidebar rail. Iconography is a custom geometric line-icon set restyled from an open-licence base such as Phosphor or Lucide — never a restrictively-licensed icon pack. Motion communicates state changes only (value ticks, connection pulses, alert entrances), capped at 200ms for UI transitions and 600ms for map/timeline scrubbing, with no decorative animation. Accessibility requirements include WCAG AA minimum contrast throughout, a colour-blind-safe alternate palette, full keyboard navigation, and both a dense "command room" large-display mode and a laptop mode.

5.2 Twenty-five screens

The interface spans 25 specified screens covering the full player and administrator lifecycle: scenario/archetype selection, country and crisis briefing, a strategic command overview with role-based field masking, an operational map (MapLibre GL JS with a custom, non-copied maritime style), a population-sentiment map, a narrative/information-environment tracker with attribution-confidence fields visible only to the Intelligence role until deliberately shared, a social/institutional network graph (Sigma.js/force-graph), an actor/organisation profile view, a shared multiplayer Cabinet Decision Room, role-specific private dashboards enforcing strict information isolation, an advisor chat interface that is retrieval-grounded and carries a visible "grounded in: state snapshot #N" indicator for trust and audit, confidence-scored intelligence reports that show intervals rather than false certainty, per-role dashboards for economics/logistics, military readiness, and diplomacy, a structured decision composer (not free chat — it outputs a schema-validated intervention object), a cross-role approval/escalation workflow, a scrubbable simulation timeline with branch-point creation, an immutable event/audit log, a no-code scenario-authoring interface (the mechanism that makes the nation-type expansion framework usable by non-engineers), a facilitator/instructor exercise-control interface for live event injection, an after-action review comparing outcomes to a Monte Carlo baseline distribution, a comparative-simulation-analysis view for side-by-side decision-branch comparison, tenant/RBAC administration, and a model/prompt/scenario version registry — the last of which is critical to the reproducibility claims made in Section 3.3.

5.3 Product-tier language discipline

Directly enforcing the brief's warning against marketing generated outcomes as real-world predictions: every AI-generated advisory text carries a visible "simulated advisory — not a real-world forecast" provenance tag, and the after-action review distinguishes simulation facts (engine-computed, reproducible) from generated narrative (LLM-authored, interpretive) at the interface level — not merely in a documentation footnote.

6. Technical Architecture and Recommended Stack

6.1 Recommended stack by layer

Layer	Recommendation	Licence	Rationale
Frontend	Next.js (React), self-hosted standalone	MIT	Richest data-visualisation ecosystem; most hireable; self-hosts cleanly for an eventual air-gapped deployment (Next.js)
Real-time sync	Socket.IO with a Redis adapter; Supabase Realtime as an optional hosted fallback	MIT / Apache-2.0	Self-hostable broadcast for tick updates and multiplayer decision rooms (Socket.IO)
Mapping	MapLibre GL JS + deck.gl (+ Turf.js)	BSD-3 / MIT	Open, self-hostable tiles, WebGL-scale rendering — avoids the Mapbox v2.0+ proprietary relicensing trap (MapLibre)

Layer	Recommendation	Licence	Rationale
Geospatial (server)	PostGIS	GPL-2.0 (as a Postgres extension)	Spatial indexing/routing colocated with the primary database
Graph visualisation	Sigma.js + graphology; react-force-graph secondary	MIT	WebGL-scale graph rendering plus social-network-analysis metrics (Sigma.js)
Backend	Python / FastAPI	MIT	Colocates with the ABM and LLM-tooling ecosystem (FastAPI)
ABM engine	Mesa + Concordia patterns for the LLM-agent layer	Apache-2.0	Proven scheduler, spatial layer, and data collection, plus a production-grade generative-agent pattern (Mesa)
Orchestration	Temporal (self-hosted); Dagster/Prefect for batch jobs	MIT	Durable, resumable, fan-out-safe simulation ticks (Temporal)
Event bus	Postgres LISTEN/NOTIFY, escalating to NATS/JetStream	PostgreSQL Licence / Apache-2.0	Right-sized for the expected scale; avoids Kafka's operational weight prematurely
Relational database	PostgreSQL	PostgreSQL Licence	The single spine — extensions collapse several other categories into one system
Graph database	Postgres + Apache AGE, escalating to Neo4j Community only if justified	Apache-2.0 / GPL-3.0	Stay on Postgres by default; escalate deliberately, with legal sign-off given Neo4j's GPL-3.0 trigger (Apache AGE)
Time-series storage	Partitioned Postgres, escalating to TimescaleDB (Apache core only)	PostgreSQL Licence / Apache-2.0	One database, one query language, direct joins to entity tables
Vector retrieval	pgvector, escalating to self-hosted Qdrant if scale demands	PostgreSQL Licence / Apache-2.0	Same database first; Qdrant only once retrieval load actually requires it
LLM gateway	LiteLLM (self-hosted)	MIT	One API surface over both local and frontier models; air-gap-ready by design (LiteLLM)
LLM providers	Local vLLM/Ollama by default; frontier APIs (OpenAI/Anthropic/etc.) only for decisions that measurably need them	—	Local inference is both the cost floor and the eventual air-gap requirement
Auth	Keycloak (self-hosted) + Postgres row-level security + OPA/Cedar	Apache-2.0	Self-hosted SSO, RBAC, and audit trail suitable for government customers (Keycloak)
Infrastructure	Container-first: Fly.io / Railway for early pilots, escalating to managed then self-hosted Kubernetes	—	Portability from a cloud pilot through to an eventual air-gapped deployment
Observability	OpenTelemetry + Grafana/Prometheus (+ Loki/Tempo)	Apache-2.0	Vendor-neutral, fully self-hostable

Layer	Recommendation	Licence	Rationale
CI/CD and infrastructure-as-code	GitHub Actions (self-hosted runners) or GitLab; OpenTofu	— / MPL-2.0	Reproducible pipelines on an OSI-open IaC tool, avoiding Terraform's BSL relicensing

6.2 Guiding principles behind the stack

Postgres functions as the spine: relational storage, PostGIS, TimescaleDB, pgvector, and Apache AGE in one system minimises the number of components that must be independently secured, backed up, and eventually re-deployed into a classified or air-gapped environment — a decisive simplification for a small team. Open-source and self-hostable components are used everywhere the eventual air-gap requirement demands it; hosted SaaS options (Abyl, Liveblocks, Pinecone, Auth0, Clerk, Datadog, Temporal Cloud, OpenRouter) are confined to optional early-pilot convenience and never sit on the critical path. LLM inference cost is the real long-run budget line: selective/tiered LLM usage, local models by default via LiteLLM, aggressive prompt caching, request batching, and "summarise-and-forget" memory management are, per production cost-engineering research, the difference between roughly \$0.50 and \$25 per agent-hour of operation ([Edgeless Lab production AI-agent cost analysis](#)). Finally, the stack deliberately rejects over-engineered components (Kafka, distributed SQL, a dedicated InfluxDB, standalone Pinecone) until a measured bottleneck at a realistic user scale (up to roughly 200 concurrent users for the pilot) actually justifies them.

6.3 Notable risk flags to track

Kùzu was abandoned by its creator in October 2025 following Apple's acquisition of the team — do not adopt it for a multi-year product regardless of its clean MIT licence ([The Register](#)). Terraform (now BSL), Memgraph, ArangoDB Community, and Redpanda Community are source-available rather than OSI-open-source, which government procurement processes often scrutinise specifically — prefer the Apache/MIT/BSD/MPL equivalents (OpenTofu, Apache AGE or carefully-isolated Neo4j, NATS) wherever the functional gap allows it.

7. AI-Assisted Development Workflow (Including What "UltraCode" Actually Is)

7.1 What "UltraCode" refers to — confirmed before designing around it

Per the brief's explicit instruction not to assume, this was verified directly: "UltraCode" is not a separate product but refers to Claude Code's high-effort, dynamic-workflow operating mode — an "Ultra Code"-style effort setting in which the coding agent runs at extra-high reasoning effort and can dynamically spawn parallel subagents to parallelise a task. The governance model below is written against Anthropic's actual, documented Claude Code best practices rather than a speculative "UltraCode" feature set.

7.2 Agent task briefing: the `CLAUDE.md` pattern

Anthropic's official guidance centres briefing on a `CLAUDE.md` file — "a special file that Claude reads at the start of every conversation," giving it "persistent context it can't infer from code alone" ([Anthropic — Best practices for Claude Code](#)). The file should stay short: "Bloated `CLAUDE.md` files cause Claude to ignore your actual instructions," and the inclusion test for any line is "would removing this cause Claude to make mistakes? If not, cut it" ([same source](#)).

Recommended contents are strictly the non-obvious items: build/test commands the agent cannot guess, code-style rules that differ from language defaults, preferred test runners, repository etiquette (branch naming, PR conventions), and — for MERIDIAN specifically — the one line that matters most: **the LLM gateway must never mutate macro or meso numeric state directly; it only returns proposals that `engine.py` validates**. This exact rule is written into the scaffold's own `CLAUDE.md`.

Anthropic frames the standard workflow as four phases — **Explore → Plan → Implement → Commit** — with a distinct "plan mode" in which the agent reads files and answers questions without making any changes, which is "most useful when you're uncertain about the approach, when the change modifies multiple files, or when you're unfamiliar with the code" ([same source](#)).

7.3 Permission boundaries

Claude Code by default requests permission for any action that might modify the system — file writes, shell commands, MCP tool calls — through three documented mechanisms: **Auto mode**, where a separate classifier model blocks only what looks risky (scope escalation, unknown infrastructure, hostile-content-driven actions); **allowlists** via `/permissions` for specific known-safe commands (e.g. `npm run lint`, `git commit`); and a **sandbox** via `/sandbox` providing OS-level filesystem and network isolation ([Anthropic — Best practices for Claude Code](#)). GitHub's parallel "safe agentic development" model for Copilot adds a layered stack of custom-instruction soft guardrails, **agent hooks** for deterministic policy enforcement, the Model Context Protocol for standardised typed tool access, and human-in-the-loop as "the final gate" ([GitHub — Safe Agentic Development with Copilot](#)).

Direct mapping to the brief's explicit prohibitions — each is a documented Claude Code or industry-standard control, not a bespoke invention:

Brief's prohibition	Enforcement mechanism
No direct production access	Sandbox mode; production credentials never loaded into the agent's environment; separate deploy identity
No unrestricted secret access	Dedicated secrets manager; agent uses scoped tokens handled in code, never exposed to the model context (OWASP LLM07 mitigation, below)
No silent dependency installation	Package managers kept <i>outside</i> the permission allowlist so <code>npm install/pip install/etc.</code> always require explicit human approval
No silent licence acceptance	A CI-gated SBOM + licence scan (Section 8.4) fails the build on any new dependency with an unreviewed licence

Brief's prohibition	Enforcement mechanism
No merging without tests	A Stop hook — deterministic, "guarantee[s] the action happens... zero exceptions" — blocks the turn from ending until the test suite passes (Anthropic)
No disabling security controls to pass tests	Security-scanning steps are themselves wired into the same Stop-hook gate, not skippable by the agent
No modification of audit logs	Signed commits, protected branches with required status checks, and an append-only audit log the agent's write identity cannot rewrite
No use of unapproved models/datasets	LiteLLM gateway allowlists approved model IDs only; a model/prompt registry (UX Screen 25) tracks what is in use
No uncontrolled shell execution	Allowlist plus sandbox plus the Auto-mode classifier, in combination
No copying code from unidentified sources	Human review requirement (below) plus provenance checks as part of the review criteria

7.4 Verification-first development and human review

Anthropic's core principle is to "give Claude a way to verify its work: tests, a build, a screenshot to compare," after which the agent "does the work, runs the check, reads the result, and iterates until the check passes" — explicitly warning against the "trust-then-verify gap" where, "without a check it can run, 'looks done' is the only signal available, and you become the verification loop" ([Anthropic](#)). Anthropic separately recommends adversarial review via "a subagent [that] review[s] the diff in a fresh context and report[s] gaps," cautioning that a reviewer prompted to find gaps will usually report some even in sound work, so it should be told to flag only gaps affecting correctness or the stated requirements ([same source](#)). Notably, Anthropic's own documentation does not mandate that a human review every change; GitHub's model, by contrast, explicitly names human-in-the-loop as "the final gate." **Reasonable inference**, given the defence-adjacent and eventual government-pilot context: mandatory human sign-off on every merge to protected branches is warranted here even though it exceeds Anthropic's strict minimum — combining (a) an AI verification subagent to catch correctness gaps with (b) a human reviewer as the final authoritative gate mirrors GitHub's stricter model and is the recommended posture for this project.

7.5 Dependency and licence guardrails, and audit-log integrity

Anthropic's documentation contains no explicit dependency-installation guardrail — the applicable controls are the general allowlist/sandbox/Auto-mode mechanisms above (**verified gap in the vendor documentation**). **Reasonable inference** to close that gap: keep package managers outside the allowlist, enforce a pre-commit/CI check that fails the build if a lockfile changes without a corresponding human approval, and run the SBOM/licence scan below as the actual review-forcing mechanism. For audit-log integrity where an agent has repository write access, use platform-side controls the agent cannot itself tamper with — signed commits (Sigstore Cosign or GPG), protected branches with required status checks, and a separation between the agent's write identity and any identity capable of rewriting history — consistent with UK NCSC guidance treating logging and monitoring as a distinct control area for the operate-and-maintain stage of an AI system's lifecycle ([NCSC — Guidelines for secure AI system development](#)).

7.6 SBOM and licence-scanning tooling

Tool	Role	Licence	Cost
Syft (Anchore)	SBOM generation from containers, filesystems, source	Apache-2.0	Free
Grype (Anchore)	Vulnerability scanning against an SBOM	Apache-2.0	Free, no seat limits
OWASP Dependency-Track	Continuous SBOM monitoring, licence/vulnerability inventory	Apache-2.0	Free, self-hosted (used by over 20,000 organisations)

The practitioner consensus is explicit: "start with Syft + Grype wired into CI/CD before you touch any commercial tool," and the full Syft-generate plus Grype-scan plus Dependency-Track-monitor open-source stack "covers generation, vulnerability scanning, and continuous monitoring at no cost" while satisfying the core machine-readable-inventory requirements of both the US Executive Order 14028 and the EU Cyber Resilience Act ([AppSec Santa — Best SBOM Tools 2026](#)). **Reasonable inference:** for a solo or small defence-adjacent team, zero-cost Syft plus Grype in CI (failing the build on high/critical CVEs) plus a single-container self-hosted Dependency-Track instance is the cost/effort-optimal posture; commercial SCA tooling is only justified later if centralized fleet dashboards or vendor support become necessary.

8. Security and Threat Model

8.1 Governing frameworks

Three complementary frameworks, used together as the sources themselves recommend: the **OWASP Top 10 for LLM Applications (2025)** for what can go wrong at the application layer ([OWASP official PDF](#)); **MITRE ATLAS**, an ATT&CK-style knowledge base of real-world adversary tactics against AI systems, explicitly designed as complementary to rather than competing with the OWASP list ([MITRE ATLAS via Vectra AI](#)); and the **NIST AI Risk Management Framework**, organised around Govern, Map, Measure, and Manage, for governance and measurement ([NIST AI RMF 1.0 PDF](#)). A companion **OWASP Multi-Agent System Threat Modeling Guide** (June 2025) is directly relevant given MERIDIAN's multi-agent architecture, and the UK NCSC / US CISA jointly-authored **Guidelines for Secure AI System Development** and the follow-on multinational **Deploying AI Systems Securely** report (co-authored with the US NSA, FBI, Australia's ACSC, Canada's CCCS, and New Zealand's NCSC-NZ) are directly applicable given the "high-threat, high-value environments" framing of the latter document ([NSA/CISA et al. PDF](#)).

8.2 Priority threats and their mitigations

Prompt injection (OWASP LLM01), unchanged as the #1 risk since 2023, splits into direct injection (user input directly alters model behaviour) and indirect injection (external content — files, scenario data — alters behaviour when parsed, with no requirement that the injected content be human-visible) ([OWASP Top 10 for LLMs v2025 PDF](#)). Documented mitigations directly applicable here: constrain model behaviour via system prompt; validate expected output formats with deterministic code rather than trusting the model's self-report; enforce least-privilege access by giving the application its own scoped API tokens handled in code rather

than exposed to the model; require human approval for high-risk actions; and segregate untrusted external content. **Reasonable inference applying this to MERIDIAN:** every scenario/config file must be treated as untrusted external content (an indirect-injection vector) and schema-validated before any agent parses it — this is precisely why the scaffold's scenario files and Pydantic schemas exist as a hard validation boundary rather than a soft convention.

Sensitive information disclosure and cross-tenant leakage (LLM02) maps directly onto MERIDIAN's asymmetric-information design (Section 4.4, Screen 10) — the same mechanism that makes the game work (role-isolated private dashboards) is also the mechanism that must be defended against context-window or session leakage across roles and tenants; enforce per-tenant and per-role isolation and specifically test for cross-context leakage.

System prompt leakage (LLM07), a new 2025 addition to the OWASP list: the definitive mitigation is to "never embed secrets... in system prompts" and treat system prompts as public, enforcing policy through separate access-control and rate-limiting layers instead ([Berezha Security Group — OWASP LLM Top 10 2025 guide](#)).

Multi-agent-specific threats, per OWASP's companion guide: "Improper Multi-Agent Trust" (agents blindly trusting outputs from other agents) and "Cross-Agent Prompt Injection" (malicious prompts propagating through inter-agent communication) are named risks specifically because MERIDIAN's institutional/adversary agents communicate with each other ([OWASP Multi-Agentic System Threat Modeling Guide v1.0](#)). Mitigation follows the same determinism-boundary principle already central to the design: no agent's output is treated as ground truth by another agent or by the engine — every inter-agent claim is validated against the deterministic state before it can have an effect.

Supply-chain vulnerabilities (LLM03) are the AI-specific analogue of ordinary code supply-chain risk and are addressed by the same SBOM stack from Section 7.6, extended to cover model weights and datasets, not only code dependencies.

8.3 Recommended combined posture

Use the OWASP LLM/Agentic Top 10 to know what can go wrong, MITRE ATLAS to test against real adversary tactics, and the NIST AI RMF to govern and measure the whole programme — as the sources themselves recommend using them together ([Vectra AI on using OWASP, ATLAS, and NIST AI RMF together](#)). Open-source LLM red-teaming tooling — **garak**, **PyRIT**, and **promptfoo** — can automate regression testing of the prompt-injection and jailbreak attack surface as the codebase evolves ([Berezha Security Group](#)).

9. Nation-Type Expansion Framework

The island-strait scenario is the version-1 demo, not a hardcoded ceiling. The architecture supports additional nation archetypes purely through scenario-configuration data — a nation type is a scenario template consisting of starting macro-indicator values, an institutional roster, a relationship-graph seed, a library of pre-authored crisis hooks, and a distinct "core tension" the archetype is built to explore. No archetype-specific code paths exist in the

simulation engine; if one seems necessary, that signals an incomplete schema, not a justified special case.

Archetype	Core tension	Population scale	Distinct systems emphasised
Small island/strait state (v1 demo)	Checkpoint control vs. great-power dependency	~2–6M	Maritime patrol, alliance triangulation, grey-zone information operations
Landlocked resource-poor state	Energy/trade dependency on neighbours	~5–15M	Transit agreements, pipeline/corridor leverage, refugee flows
Oil-rich Gulf-style state	Resource wealth vs. succession/legitimacy risk	~5–10M	Sovereign wealth fund allocation, rentier-state social contract, OPEC-style coordination
Regional/European mid-power	Alliance-bloc obligations vs. domestic politics	~40–80M	Coalition fragility, EU/NATO-style multilateral constraints, energy transition
Global superpower	Overextension vs. domestic legitimacy	300M+	Multi-theatre resource allocation, proxy-conflict management, reserve-currency/financial-warfare tools
Failed/fragile state (playable adversary-adjacent)	Monopoly-on-violence collapse	~10–30M	Warlord/militia agent networks, humanitarian-actor interplay, foreign-intervention triggers
Federated/multi-ethnic state	Centre-periphery legitimacy	~20–50M	Devolution mechanics, secessionist cohort modelling, constitutional crisis paths

What actually changes per archetype, all data-driven: macro starting values (e.g. a superpower archetype starts with high military readiness but lower approval elasticity; an oil-rich archetype starts with high treasury reserves but a single point of failure in FDI/oil-price sensitivity); the institutional roster (a Gulf-style archetype adds a royal-court/succession-council institutional agent; a superpower archetype adds joint chiefs, a central-bank equivalent, and multiple regional commands); the relationship-graph topology (hub-and-spoke for a superpower versus triangulation for a small state); a crisis-hook library of three to five pre-authored seeds per archetype using the same event/campaign schemas as the island demo; and a tuned victory/failure condition set appropriate to that archetype's core tension.

The selection flow mirrors the ChatGPT-scenario-game pattern that originally inspired this project: at scenario start, the player chooses an archetype from a card-based selector (UX Screen 1), each card showing a one-line tension description and a difficulty indicator; the chosen archetype loads its scenario-template JSON into the same generic engine — adding an eighth archetype requires zero new engine code, only a new template file conforming to the schemas already defined in Section 3. Multiplayer roles are similarly archetype-agnostic and simply relabelled per scenario (e.g. "Head of State" becomes "President," "Prime Minister," "Chairman," or "Emir" depending on the loaded scenario's government-structure field), with identical underlying permission, private-information, and decision-composer systems across every archetype.

10. Starter Repository Scaffold

A runnable scaffold accompanies this document at the path noted in the delivery message, implementing the architecture above rather than merely describing it. It was built, executed, and independently re-verified: **5 of 5 tests pass** (`test_runs_a_few_ticks`, `test_same_seed_is_deterministic`, `test_different_seed_diverges`, `test_macro_state_actually_changes`, `test_llm_gateway_cannot_write_state`), confirmed by re-running the suite directly rather than trusting the build subagent's self-report.

What it contains: a FastAPI backend with Pydantic v2 schemas matching Section 3's data model field-for-field (with per-field descriptions), nine JSON Schema mirrors of those same objects for cross-language tooling, a Mesa-based simulation engine implementing the macro/meso/micro tiers as a `MeridianModel`, a seeded Linear-Threshold diffusion module operating over a `NetworkX` cohort graph, an LLM-gateway stub whose function signatures make it structurally impossible to write macro state directly (it can only return `ActionProposal` objects with no reference to the state object at all), REST and WebSocket API routes, SQLAlchemy persistence models for simulation runs, state snapshots, and the event log, a `docker-compose.yml` bringing up Postgres alongside the backend, the full "Kestral Strait" demo scenario as structured JSON data (five cohorts, six institutional roles, the hidden camp-eastern-veterans-01 campaign, and win/loss/ambiguous conditions), nine architecture-decision records, an AI-coding-agent task-briefing template, and a project-level `CLAUDE.md` encoding the non-negotiable rule from Section 3.2.

How to run it: `docker-compose up` brings up Postgres; `pip install -r backend/requirements.txt` followed by `python -m pytest backend/tests` from the `scaffold/backend` directory runs the verification suite with no API keys required, since the LLM gateway operates in a stub/mock mode by default.

Intended use: this is a skeleton for an AI coding agent (or a small team using one) to build outward from — it deliberately does not attempt a complete product. The determinism and LLM-isolation properties are the two things that must never regress as the codebase grows, which is why both are enforced by tests rather than only by convention.

11. Phased Build Roadmap

Phase 0 — Demo (current deliverable + immediate next steps). Ship the scaffold's Kestral Strait scenario as a single-player, local-only playable demo: connect the existing REST/WebSocket routes to a minimal frontend (the scaffold's stub is a placeholder, not the Screen 1–25 UX system), wire a real LLM provider behind the LiteLLM gateway for advisor dialogue and action proposals (local model first, per the cost-engineering guidance in Section 6.2), and validate the core determinism/isolation properties hold under actual LLM-driven play, not just synthetic test data.

Phase 1 — Pilot. Build out the multiplayer Cabinet Decision Room and role-isolated private dashboards (the asymmetric-information mechanic that makes the game work), stand up the full 25-screen UX system against the design tokens in Section 5, add the second nation archetype (recommended: the oil-rich Gulf-style state, since its succession/legitimacy tension

is maximally distinct from the strait scenario and will stress-test whether the "no archetype-specific code" principle actually holds), and put the CI-gated SBOM/licence-scanning and human-review-required merge gates from Section 7 into force before any external pilot user touches the system.

Phase 2 — Air-gapped/government-ready. Complete the migration path already implicit in every "Recommended" stack choice in Section 6 — every component was chosen specifically because it self-hosts — by validating a fully offline deployment (local model inference only, no external API calls, no SaaS dependencies on the critical path), formalising the Keycloak-based RBAC and audit-log integrity controls from Section 7.5 for a compliance audit, and commissioning the legal review flagged in Section 2.2 for any BUSL/BSL component still in use at that point (ideally, none — the recommended default stack in Section 2.3 avoids them entirely).

Each phase gate is a go/no-go decision point, not an assumed progression — the pilot should validate real user engagement with the asymmetric-information multiplayer mechanic before any investment in the air-gapped phase, since that mechanic (not the LLM, not the visual polish) is the actual product hypothesis being tested.